

After Simon, after Hrushovski

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ABSTRACT.

1. Patterns

In the following $\mathcal{U}$ is some monster model of a complete theory T of language L . For simplicity T is assumed to be one-sorted.

We define a new structure $\mathcal{N}$ whose domain is a subset of $\mathcal{U}^x$, where x is some fixed tuple of variables. The language of $\mathcal{N}$ is denoted by L^p . Though L^p depends on x , this will be left implicit. We also use an auxiliary tuple $z = \langle z_i : i < \lambda \rangle$ of (so-called) parameter variables.

Fix a set of formulas $L_z \subseteq \Delta(x; z) \subseteq L_{x,z}$. Let V be some large set of variables that are copies of x . Finally, let Δ be the set of positive Boolean combinations of formulas in $\bigcup_{x' \in V} \Delta(x'; z)$. The choice of Δ is justified in Remark 1.

Let $\bar{a}, \bar{a}' \in (\mathcal{U}^x)^n$ and $B \subseteq \mathcal{U}$. If $\varphi(\bar{a}; b) \leftrightarrow \varphi(\bar{a}'; b)$ holds for every $\varphi(\bar{x}, z) \in \Delta$ and $b \in B^z$, we write $\bar{a} \equiv_{\Delta, B} \bar{a}'$. Let $b, b' \in \mathcal{U}^z$ and $A \subseteq \mathcal{U}$. We write $b \equiv_{A, \Delta} b'$ if $\varphi(\bar{a}; b) \leftrightarrow \varphi(\bar{a}; b')$ holds for every $\varphi(\bar{x}, z) \in \Delta$ and $\bar{a} \in A^{\bar{x}}$.

For every formula $\varphi(x_0, \dots, x_{n-1}; z) \in \Delta$ and every formula $\alpha(z) \in L$ the language L^p contains the n -ary relation $r_{\varphi; \alpha}$. Below by the symbol L^p we will also denote the set of *atomic* formulas $r_{\varphi; \alpha}(x_0, \dots, x_{n-1})$ for any $x_0, \dots, x_{n-1} \in V$. Note that equalities are *not* included in L^p .

Given pairs of models $M \leq N$, where N is $|M|^+$ -saturated, we define a model $\mathcal{N}$ with domain N^x . We define the interpretation of $r_{\varphi; \alpha}$ in $\mathcal{N}$. If $\bar{a} = a_0, \dots, a_{n-1} \in \mathcal{N}$, we declare $r_{\varphi; \alpha}(\bar{a})$ true if $\varphi(\bar{a}; b)$ holds for every $b \in M^z$ such that $\alpha(b)$. We repeat the definition for emphasis

$$r_{\varphi; \alpha}(\bar{a}) \Leftrightarrow \alpha(M^z) \subseteq \varphi(\bar{a}; M^z).$$

The following is paramount.

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1 Remark By the choice of Δ , the truth of $r_{\varphi;\alpha}(\bar{a})$ only depends on the equivalence class of $\equiv_{\Delta,M}$ of the *components* of the tuple $\bar{a}$. Explicitely, if $\bar{a} = a_0, \dots, a_{n-1}$ and $\bar{a}' = a'_0, \dots, a'_{n-1}$ are such that $a_i \equiv_{\Delta,M} a'_i$ for every $i < n$, then

$$\# \quad r_{\varphi;\alpha}(\bar{a}) \leftrightarrow r_{\varphi;\alpha}(\bar{a}').$$

By the remark, it would be natural to define the interpretation of L^P on the quotient $\mathcal{N}/\equiv_{\Delta,M}$. But we prefer to keep the original domain $\mathcal{N}$ to have a common playground with L .

A **p-type** is a set of (atomic) formulas in $L^P(\mathcal{N})$. We write **p-tp**($\bar{a}$) for the p-type of $\bar{a}$, i.e. the set of formulas $r_{\varphi;\alpha}(\bar{x})$ such that $r_{\varphi;\alpha}(\bar{a})$ holds.

Note incidentally that the relation $x_0 \equiv_{\Delta,M} x_1$ is definable by a p-type. Namely the p-type containing the formulas $r_{\vartheta;\top}(x_0, x_1)$ for every $\vartheta(x_0, x_1; z)$ of the form $\varphi(x_0; z) \leftrightarrow \varphi(x_1; z)$.

2 Fact Formulas in L^P are closed under adjunction of dummy variables, disjunction, conjunction, and existential quantification.

Proof. The claim about dummy variables is clear, so we tacitly use it below.

Up to renaming of the variables that occur in $\beta(z)$ and $\psi(\bar{x}; z)$, we can assume that they are disjoint from those in $\alpha(z)$. Therefore the following equivalences hold

$$\begin{aligned} r_{\varphi;\alpha}(\bar{a}) \vee r_{\psi;\beta}(\bar{a}) &\Leftrightarrow \forall z \in \alpha(M^z) \varphi(\bar{a}; z) \vee \forall z \in \beta(M^z) \psi(\bar{a}; z) \\ &\Leftrightarrow \forall z \in \alpha(M^z) \cap \beta(M^z) \varphi(\bar{a}; z) \vee \psi(\bar{a}; z) \\ &\Leftrightarrow r_{\varphi \vee \psi; \alpha \cap \beta}(\bar{a}) \end{aligned}$$

A similar argument applies to the conjunction of two formulas.

Finally, consider the formula $\exists y r_{\varphi;\alpha}(\bar{x}, y)$. Without loss of generality, we can assume that φ has the form

$$\varphi(\bar{x}, y; z) = \bigvee_{i=1}^m \varphi_i(\bar{x}, y; z)$$

where $\varphi_i(\bar{x}, y; z) = \varphi'_i(\bar{x}; z) \wedge \psi_i(y; z)$ and each $\varphi'_i(\bar{x}; z)$ is a conjunction of formulas in $\bigcup_{x' \in V} L_{x'; z}$. Closure under existential quantification follows from the following equivalences

$$\begin{aligned} \exists y r_{\varphi;\alpha}(\bar{a}, y) &\Leftrightarrow \exists y \forall z \left[z \in \alpha(M) \rightarrow \bigvee_{i=1}^m \varphi_i(\bar{a}, y; z) \right] \\ &\Leftrightarrow \forall z \left[z \in \alpha(M) \rightarrow \bigvee_{i=1}^m \varphi'_i(\bar{a}; z) \wedge \exists y \psi_i(y; z) \right] \\ &\Leftrightarrow r_{\vartheta;\alpha}(\bar{a}), \end{aligned}$$

where

$$\vartheta(\bar{x}; z) = \bigwedge_{i=1}^m \varphi'_i(\bar{x}; z) \wedge \exists y \psi_i(y; z)$$

is a positive Boolean combination of formulas in $\bigcup_{x' \in V} L_{x'; z}$, as required. $\square$

3 Fact (∞ -p-saturation) Let $A \subseteq \mathcal{N}$ have arbitrary cardinality. Let $\bar{x}$ be a tuple of copies of x of length $\leq |\mathcal{N}|$. Then every p-type $p(\bar{x}) \subseteq L^p(A)$ that is finitely consistent in $\mathcal{N}$ is realized in $\mathcal{N}$.

Proof. Let $A' \subseteq \mathcal{N}$ be a small set containing a representative of every $\equiv_{\Delta, M}$ equivalence class that intersects A . Replacing the elements of A that occur in $p(\bar{x})$ by elements of A' we obtain an equivalent type $p'(\bar{x}) \subseteq L^p(A')$. Note that every formula $r_{\varphi; \alpha}(\bar{x})$ there is a Δ -type $q(\bar{x}) \subseteq L(M)$ such that

$$\mathcal{N} \models r_{\varphi; \alpha}(\bar{a}) \Leftrightarrow \mathcal{N} \models q(\bar{a})$$

Therefore the saturation of $\mathcal{N}$ implies that $p'(\bar{x})$ is realized in $\mathcal{N}$. $\square$

2. Morphism

A p-morphism is a map $f : \mathcal{N} \rightarrow \mathcal{N}$ such that for every $r_{\varphi; \alpha}(\bar{x}) \in L^p$ and every $\bar{a} \in (\text{dom } f)^{\bar{x}}$

$$\# \quad \mathcal{N} \models r_{\varphi; \alpha}(\bar{a}) \Rightarrow \mathcal{N} \models r_{\varphi; \alpha}(f\bar{a}).$$

As $\equiv_{\Delta, M}$ is definable by a p-type, p-morphisms induce well-defined maps on the quotient $\mathcal{N}/\equiv_{\Delta, M}$. We say that two p-morphisms f and g are **equivalent**, and write $f \sim g$, if they induce the same map on $\mathcal{N}/\equiv_{\Delta, M}$. We often confuse equivalent p-morphisms and we use the same symbol f to denote the map induced on the quotient.

A total p-morphism is called a p-endomorphism. We write $\text{End}(\mathcal{N})$ for the set of p-endomorphisms of $\mathcal{N}$. We endow $\text{End}(\mathcal{N})$ with the topology that has as basic closed sets those of the form

$$\left\{ f \in \text{End}(\mathcal{N}) : \mathcal{N} \models r_{\varphi; \alpha}(fa_1, \dots, fa_n, c_1, \dots, c_m) \right\}$$

for $\varphi(x; z) \in \Delta$, $\alpha(z) \in L$, and $a_1, \dots, a_n, c_1, \dots, c_m \in \mathcal{N}$. Note that, by Remark 1, if $f \sim g$ then f and g belong to the same basic closed sets, therefore this topology naturally induces a topology on $\text{End}(\mathcal{N})/\sim$.

4 Fact With the above topology $\text{End}(\mathcal{N})$ is T_1 (singletons are closed) and compact. Composition of p-morphisms is a left continuous operation.

Proof. We prove that singletons are closed. In fact

$$\{f\} = \bigcap_{a \in \mathcal{N}} \left\{ g \in \text{End}(\mathcal{N}) : a \equiv_{\Delta, M} ga \right\}$$

therefore, as $\equiv_{\Delta, M}$ is defined by a p-type, $\{f\}$ is the intersection of closed sets.

We prove compactness. Assume the following intersection is finitely consistent (we have slightly compressed the notation)

$$\#. \quad \bigcap_{i < \lambda} \left\{ f \in \text{End}(\mathcal{N}) : \mathcal{N} \models r_i(f\bar{a}, \bar{c}) \right\}.$$

then the p-type

$$\bar{x} \equiv_{\Delta, M} \bar{a} \cup \{r_i(\bar{x}, \bar{c}) : i < \lambda\}$$

is finitely consistent in $\mathcal{N}$. By ∞ -p-saturation, it is realized by some $\bar{a}'$. By Fact 7 there is a p-endomorphism $\mathcal{N}$ that maps $\bar{a}$ to $\bar{a}'$. This proves that the set in # is non empty.

We prove left continuity of composition. That is for any given $f \in \text{End}(\mathcal{N})$ we prove that the function $\tau_f : \text{End}(\mathcal{N}) \rightarrow \text{End}(\mathcal{N})$ that maps $g \mapsto g \circ f$ is continuous. Pick any basic closed set $\{h : r(h\bar{a}, \bar{c})\}$. Then

$$\tau_f^{-1}[\{h : r(h\bar{a}, \bar{c})\}] = \{g : r(g(f\bar{a}), \bar{c})\}$$

is closed, as required. $\square$

If R is a p-morphism, also $\equiv_M \circ R \circ \equiv_M$ is a p-morphism. This we call the **full** p-morphism generated by R . We say that R and R' are **equivalent** if they generate the same full p-morphism.

5 Remark We can identify p-morphisms with maps $f : \mathcal{U}^x / \equiv_M \rightarrow \mathcal{U}^x / \equiv_M$. In fact, to any such map we can associate the relation $R = \bigcup_{a \in \mathcal{U}^x} ((a / \equiv_M) \times f(a / \equiv_M))$.

Vice versa, to any p-morphisms R we can associate $f : \mathcal{U}^x / \equiv_M \rightarrow \mathcal{U}^x / \equiv_M$ that maps $a / \equiv_M \mapsto b / \equiv_M$ for some/any b such that $\langle a, b \rangle \in R$. This is well-defined because if $\langle a, b \rangle, \langle a', b' \rangle \in R$ and $a \equiv_M a'$, then we necessarily have $b \equiv_M b'$.

We find more convenient to work with relations.

We say that the p-morphism R is **reduced** if there is no $R' \subset R$ that is equivalent to R . It is clear that if $A, B \subseteq \mathcal{U}^x$ are any two sets of representative of $\equiv_M$ then $f = R \cap (A \times B)$ is a reduced morphism equivalent to R .

6 Fact Any reduced p-morphism is the graph of a (partial) map. Therefore every reduced p-isomorphism is a bijection.

Proof. Suppose not, then there are a, b, c such that $\langle a, b \rangle, \langle a, c \rangle \in R$ but $b \neq c$. As R is reduced, then $b \not\equiv_M c$. As noted above, this is not possible. The second claim follows immediately. $\square$

7 Fact Every p-morphism extends to a p-morphism defined on the whole of $\mathcal{U}^x$. Also, every p-isomorphism extends to a p-automorphism of $\mathcal{U}^x$.

Proof. Let R be a p-morphism. Pick be a reduced p-morphism f equivalent to R . Let $\bar{a}$ be an enumeration of $\text{dom } f$ and let $b \in \mathcal{U}^x$. Let $p(\bar{x}) = \text{p-tp}(\bar{a})$ and let $q(\bar{x}, y) = \text{p-tp}(\bar{a}, b)$. By ∞ -p-saturation $\exists y q(\bar{x}, y)$ is equivalent to the type containing the existential closure of the formulas in $q(\bar{x}, y)$. By Fact 2, the latter is equivalent to a p-type, which is necessarily coincides with $p(\bar{x})$. Then $q(f\bar{a}, y)$ is consistent hence by ∞ -p-saturation

it is realized by some $c \in \mathcal{U}^x$. Then $f \cup \{\langle b, c \rangle\}$ is a p-morphism defined in b that extends f . Then R can be extended as well.

The second claim is obtained by back-and-forth. $\square$

3. Duality

In this section $\hat{m}$ is an enumeration of M and $|z| = |\hat{m}|$. For a more convenient notation, x is taken of infinite length – then, up to equivalence, formulas in $L_z(\mathcal{U})$ have the form $\varphi(a; z)$ for some $\varphi(x; z) \in L$ and $a \in \mathcal{U}^x$.

8 Remark We stress a simple fact which is used in the proof below. We say that $\varphi'(a; z)$ is an alphabetic variant of $\varphi(a; z)$ if it is obtained by replacing the variables effectively occurring in $\varphi(a; z)$ by other variables among those in z . The following are clearly equivalent

1. $\varphi(a; M^z) = M^z$
2. $\varphi'(a; \hat{m})$ holds for every alphabetic variant of $\varphi(a; z)$.

Let $\mathcal{B} = \{b \in \mathcal{U}^z : \hat{m} \equiv b \downarrow_M N\}$, where $\equiv$ is the elementary equivalence in L however, as $L_z \subseteq \Delta$, we could replace it with $\equiv_{\emptyset, \Delta}$. The N, Δ -topology on $\mathcal{B}$ is the one generated by the closed sets of the form $\{b \in \mathcal{B} : \varphi(\bar{a}; b)\}$ for $\varphi(\bar{x}; z) \in \Delta$ and $\bar{a} \in \mathcal{N}^{\bar{x}}$. The N, Δ -topology on $\mathcal{B}/\equiv_{N, \Delta}$ is the quotient of that on $\mathcal{B}$.

9 Fact The topological spaces $\text{End}(\mathcal{N})/\sim$ and $\mathcal{B}/\equiv_{N, \Delta}$ are homeomorphic.

Proof. In 1 below, we define a map $f \mapsto b_f$ from $\text{End}(\mathcal{N})$ into $\mathcal{B}$. In 2, we define a map $b \mapsto f_b$ from $\mathcal{B}$ into $\text{End}(\mathcal{N})$. The reader may check that these maps induce maps between $\text{End}(\mathcal{N})/\sim$ and $\mathcal{B}/\equiv_{N, \Delta}$ that are each other's inverse. In 3 we prove the continuity of the first map and in 4 the continuity of the second.

1. Let $f : \mathcal{N} \rightarrow \mathcal{N}$ be an p-epimorphism. Let $p(z)$ be the type containing the formulas $\varphi(a; z)$ for $\varphi(x; z) \in \Delta$ and $a \in \mathcal{N}$ such that $\varphi(fa; \hat{m})$. Clearly $p(z)$ contains $p_{\hat{m}}(z)$ and is complete over N, Δ . We prove that $p(z)$ is finitely satisfied in M , then b_f is some/any realization of $p(z)$.

If for a contradiction that $p(z)$ is not finitely satisfied in M , then $\psi(a_1, \dots, a_n; M^z) = \emptyset$, where

$$\psi(x_1, \dots, x_n; z) = \bigwedge_{i=1}^n \varphi_i(x_i; z)$$

for some $\varphi_i(x; z) \in L$ and $a_i \in \mathcal{N}$ such that $\varphi_i(fa_i; \hat{m})$ for every i . Then $r_{\neg\psi; \top}(a_1, \dots, a_n)$ and $\neg r_{\neg\psi; \top}(fa_1, \dots, fa_n)$. This is not possible because f is a p-morphism.

2. Let $b \in \mathcal{B}$ be given. We define

$$p(x, y, z) = \{\varphi(y; z) \rightarrow \varphi(x; \hat{m}) : \varphi(x; z) \in \Delta\}.$$

Note that $p(x, a, b)$ is a p-type, for any $a \in \mathcal{N}$ and $b \in \mathcal{B}$. We claim that if $a \equiv_{\Delta, M} a'$ then $p(x, a, b) = p(x, a', b)$. In fact, $\varphi(a; b) \leftrightarrow \varphi(a'; b)$ would contradict $b \perp_M N$. Also, note that $p(x, a, b)$ is consistent, otherwise $\neg \exists x \wedge_i \varphi_i(x; \hat{m})$, for some finitely many formulas $\varphi_i(x; z) \in \Delta$ such that $\varphi_i(a; b)$. But this would contradict $b \equiv \hat{m}$.

Then we can define $f_b(a)$ to be some/any realization of $p(x, a, b)$.

We prove that f_b is a p-morphism. Let $\psi(x_1, \dots, x_n; z) \in \Delta$, $\alpha(z) \in L$, and $a_1, \dots, a_n \in \mathcal{U}^x$ be such that $r_{\psi; \alpha}(a_1, \dots, a_n)$. Assume

$$\psi(x_1, \dots, x_n; z) = \bigvee_{h=1}^m \bigwedge_{k=1}^m \varphi_{h,k}(x_{i_{h,k}}; z)$$

for some $i_{h,k} \in \{1, \dots, n\}$. Now, after re-indexing, from $r_{\psi; \alpha}(a_1, \dots, a_n)$ we obtain that $\alpha(M^z) \subseteq \varphi_k(a_k; M^z)$ for $k = 1, \dots, m$. Then from $b \perp_M N$ we obtain that b satisfies $\alpha(z) \rightarrow \varphi_k(a_k; z)$ and all its alphabetic variants. Therefore $\hat{m}$ satisfies $\alpha(z) \rightarrow \varphi_k(f_b a_k; z)$ and all its alphabetic variants. Therefore $\alpha(M^z) \subseteq \varphi_k(f_b a_k; M^z)$ and $r_{\psi; \alpha}(f_b a_1, \dots, f_b a_n)$ follows.

3. We prove that the map defined in 1 is continuous. Pick any basic closed set in $\mathcal{B}$, say $\{b : \varphi(\bar{a}; b)\}$. We claim that the set $\{f : \varphi(\bar{a}; b_f)\}$ is closed in $\text{End}(\mathcal{N})$. Indeed, by the definition of b_f in 1

$$\begin{aligned} \{f : \varphi(\bar{a}; b_f)\} &= \{f : \varphi(f\bar{a}; \hat{m})\} \\ \# &= \{f : r_{\psi; \top}(f\bar{a}, c)\} \end{aligned}$$

where $\psi(\bar{x}, y; z)$ and c are choosen so that $r_{\psi; \top}(\bar{x}, c)$ is equivalent to $\varphi(\bar{x}; \hat{m})$. As the set in $\#$ is closed in $\text{End}(\mathcal{N})$, continuity of the map $f \mapsto b_f$ follows.

We show how to choose $\psi(\bar{x}, y; z)$ and c . Let y be a copy of x . The formula $\psi(\bar{x}, y; z)$ is obtained from $\varphi(\bar{x}; z)$ by replacing the variables in the tuple z with variables in the tuple y . Note that z is dummy in $\psi(\bar{x}, y; z)$. Finally, we choose $c \in M^y$ such that $\psi(\bar{x}, c; z)$ is equivalent to $\varphi(\bar{x}; \hat{m})$.

4. We prove that the map defined in 2 is continuous. Pick any basic closed set in $\text{End}(\mathcal{N})$, say $\{h : r(h\bar{a}, \bar{c})\}$. We claim that the set $\{b : r(f_b \bar{a}, \bar{c})\}$ is closed. Indeed, by the definition of f_b in 2,

$$\{b : r(f_b \bar{a}, \bar{c})\} = \left\{ b : b \models \exists \bar{x} \left[\bigwedge_{i=1}^n p(x_i, a_i, b) \wedge r(\bar{x}, \bar{c}) \right] \right\}$$

is closed in $\mathcal{B}$ because $r(\bar{x}, \bar{c})$ is definable by a N, Δ -type. $\square$

We define a semigroup operation on $\mathcal{B}$. Namely, if $f_b, f_c : \rightarrow$ are the p-morphisms the correspond to $b, c \in \mathcal{B}$ we write $b * c$ for the element of B that corresponds to the p-morphism $f_b \circ f_c$.

4. Semigroups

Assume $\mathcal{U}$ is a semigroup w.r.t. some definable operation which we denote by $\cdot$. In this section $|x| = 1$. We could assume that also $|z| = 1$ but, we prefer to stick to the very same

tuple of variables $z = \langle z_i : i < \lambda \rangle$ used above. We may write $z \cdot x$ for the tuple $z \cdot x$. In this section $\Delta(x, z)$ contains L_z and the formulas $\varphi(z \cdot x)$ for all $\varphi(x) \in L$.

10 Fact Let $b \in \mathcal{B}$ be given. Let $f : \mathcal{N} \rightarrow \mathcal{N}$ be the function that maps $a \mapsto b \cdot a$. Then f is a p-morphism.

Proof. We prove that f is well-defined. Pick any two $a \equiv_{\Delta, M} a'$ in $\mathcal{N}$. As $b \perp_M \mathcal{N}$ then $a \equiv_{\Delta, M, b} a'$, it follows that $b \cdot a \equiv_{\Delta, M} b \cdot a'$ as required.

Now, assume $r_{\psi, \alpha}(a_1, \dots, a_n)$. We prove $r_{\psi, \alpha}(fa_1, \dots, fa_n)$. Assume that for some $i_{h,k} \in \{1, \dots, n\}$

$$\psi(x_1, \dots, x_n; z) = \bigvee_{h=1}^m \bigwedge_{k=1}^m \varphi_{h,k}(z \cdot x_{i_{h,k}}) \wedge \varphi'_{h,k}(z)$$

for some $\varphi_{h,k}(z_0; z), \varphi'_{h,k}(z) \in \Delta$.

From $r_{\psi, \alpha}(a_1, \dots, a_n)$, after re-indexing, we obtain that $\alpha(M^z) \subseteq \varphi_k(M^z \cdot a_k) \cap \varphi'_k(M^z)$ for $k = 1, \dots, m$. As $b \perp_M \mathcal{N}$, we immediately obtain $\alpha(M^z) \subseteq \varphi_k(M^z \cdot b \cdot a_k)$ for all k , so $r_{\psi, \alpha}(fa_1, \dots, fa_n)$ follows. $\square$

5. Cores

A set $A \subseteq \mathcal{N}$ is **p-maximal** if it satisfies the following (obviously) equivalent conditions for every $\bar{a} \in A$ and every p-morphism f

- a. $r_{\varphi; \alpha}(\bar{a}) \Leftrightarrow r_{\varphi; \alpha}(f\bar{a})$ for every $r_{\varphi; \alpha} \in L^p$
- b. $\text{p-tp}(\bar{a}) = \text{p-tp}(f\bar{a})$.

11 Fact The following are equivalent

1. $A \subseteq \mathcal{N}$ is p-maximal
2. $p(\bar{x}) = \text{p-tp}(\bar{a})$ is a maximally consistent p-type for every $\bar{a} \in A^{\bar{x}}$ and every finite $\bar{x}$
3. $p(\bar{y}) = \text{p-tp}(\bar{c})$ is a maximally consistent p-type for $\bar{c}$ any enumeration of A
4. every p-morphism defined on A is a p-isomorphism between A and its image.

Proof. $1 \Rightarrow 2$. Negate 2. Suppose $\text{p-tp}(\bar{a})$ is not maximal for some finite $\bar{a} \in A^{\bar{x}}$. Pick some $\bar{a}' \in A^{\bar{x}}$ such that $\text{p-tp}(\bar{a}) \subset \text{p-tp}(\bar{a}')$. Then $f = \{\langle \bar{a}, \bar{a}' \rangle\}$ is a p-morphism that contradicts the p-maximality of A .

$2 \Rightarrow 3$. If $\bar{c}$ is an enumeration of A that witnesses the failure of 3, then some finite subtuple of $\bar{c}$ witnesses the failure of 2.

$3 \Rightarrow 4$. Negate 4. Let f be a p-morphism defined on A that is not a p-isomorphism. Let $\bar{a} \in A^{\bar{x}}$ and $\bar{b}$ be such that $\langle \bar{a}, \bar{b} \rangle \in R$ and $\text{p-tp}(\bar{a}) \setminus \text{p-tp}(\bar{b}) \neq \emptyset$. To contradict 3, it suffices to extend $\{\langle \bar{a}, \bar{b} \rangle\}$ to a p-morphism defined on A . This is possible by ∞ -p-homogeneity.

$4 \Rightarrow 1$. Clear. $\square$

12 Example Assume that $a \in \mathcal{U}^x$ is such that $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$ for every $\varphi(x; z) \in L$. Then the orbit of a under $\text{Aut}(\mathcal{U}/\{M\})$ is p-maximal.

Proof. First consider the case when $\varphi(a; M^z)$ is definable over $\emptyset$. We prove that $\{a\}$ is p-maximal.

Let $\varphi'(z) \in L$ be such that $\varphi(a; M^z) \leftrightarrow \varphi'(M^z)$. Then, if we set $\varphi''(x; z) = \varphi(x; z) \leftrightarrow \varphi'(z)$, the formula $r_{\varphi''; \top}(a)$ holds.

If $\{\langle a, b \rangle\}$ is a p-morphism then $r_{\varphi''; \top}(b)$ also holds. Then $a \equiv_{\Delta, M} b$ follows. This shows that p-morphism are trivial and therefore p-isomorphisms.

Now assume that $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$. ??? □

A set $\mathcal{J} \subseteq \mathcal{U}^x$ is called a **core** if it is p-maximal and, among the p-maximal sets, it is maximal w.r.t. inclusion.

13 Fact Let $\mathcal{J}$ be a core. Let $\mathcal{A}$ be a p-maximal set. Then every p-morphism $R \subseteq \mathcal{A} \times \mathcal{J}$ extends to a p-morphism R' such that $\mathcal{A} = \text{dom} R'$ and $\text{range} R' \subseteq \mathcal{J}$.

Proof. Reason as in the proof of Fact 7 with the same notation but choose $b \in \mathcal{A}$. Let R' be a p-morphism defined on $\mathcal{A}$ that extends $f \cup \{\langle b, c \rangle\}$. Then $R'[\mathcal{A}]$, i.e. the image of $\mathcal{A}$ under R' , is a p-maximal set that contains $\mathcal{J}$. By maximality $R'[\mathcal{A}]$ coincides with $\mathcal{J}$, hence $c \in \mathcal{J}$. □

By back-and-forth we obtain the following.

14 Corollary (∞ -p-homogeneity of the core) If $\mathcal{J}$ and $\mathcal{J}'$ are cores, then every p-morphism $R \subseteq \mathcal{J} \times \mathcal{J}'$ extends to a p-isomorphism between $\mathcal{J}$ and $\mathcal{J}'$.

A **p-retraction** of $\mathcal{U}^x$ onto $\mathcal{A}$ is a total p-morphism R that coincides with $\equiv_{\Delta, M}$ on $\mathcal{A}$, i.e. such that $R \cap (\mathcal{A} \times \mathcal{A}) = (\equiv_{\Delta, M}) \cap (\mathcal{A} \times \mathcal{A})$.

15 Theorem If $\mathcal{J}$ and $\mathcal{J}'$ are cores then there is a p-retraction of $\mathcal{U}^x$ onto $\mathcal{J}$

Proof. Let $\bar{c}$ enumerate $\mathcal{U}^x$. Let $\bar{a}$ be a tuple such that $\text{p-tp}(\bar{a}) \supseteq \text{p-tp}(\bar{c})$ is p-maximal. By Fact ?? the relation $R = (\equiv_M \circ \{\langle \bar{c}, \bar{a} \rangle\} \circ \equiv_M)$ is a p-morphism onto the p-maximal set $\mathcal{A} = \text{range } R$. As $\mathcal{A}$ is p-maximal, by Fact 13 we can extend $(R \cap (\mathcal{J} \times \mathcal{A}))^{-1}$ to a p-isomorphism R' defined on $\mathcal{A}$. It is easy to verify that $R' \circ R$ is a retraction. □

16 Fact The following are equivalent

1. $a \in \bigcap \{\mathcal{J} : \mathcal{J} \text{ is a core}\}$
2. $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$, for every formula $\varphi(x; z) \in L$.

Proof. ??? □

17 Definition Let $\mathcal{J}$ be a core; we define a topology on $\mathcal{J}$. The closed sets in the p-topology on $\mathcal{J}$ are those defined by p-types with (arbitrary many) parameters in $\mathcal{J}$.

18 Fact The p-topology is compact and T_1 .

Proof. ???

□

6. Esperimenti

Let M be a saturated model and let $G = \text{Aut}(M)$.

The group G acts on $\mathcal{U}^x / \equiv_M$ in the natural way.